



Air pollution from industries and asthma onset in childhood: A population-based birth cohort study using dispersion modeling

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ABSTRACT

Background: Despite evidence that ambient air pollution may play a role in the development of asthma, little is known about the potential contribution of industrial emissions.

Objective: We used a population-based birth cohort to investigate the association between asthma onset in childhood and residential exposure to industrial emissions, estimated from atmospheric dispersion modeling.

Methods: The study population comprised all children born in the province of Quebec, Canada, 2002–2011. Asthma onset were ascertained from health administrative databases with validated algorithms. We used atmospheric dispersion modeling to develop time-varying annual mean concentration of ambient PM_{2.5}, NO₂ and SO₂ at participants' residence from industries. For each pollutant, we assessed the association between industrial emissions exposure and childhood asthma onset using Cox proportional hazard model, adjusted for sex, material and social deprivation and calendar year. Sensitivity analysis included adjusting for long-term regional and traffic-related ambient PM_{2.5} and NO₂, and assessing potential confounding by unmeasured secondhand smoke.

Results: The cohort included 722,667 children and 66,559 incident cases of asthma. For all pollutants, we found a non-linear association between childhood asthma onset and residential ambient air pollutant concentration from industries, with stronger effects at lower concentrations. A change from 25th to the 75th percentile in the mean annual ambient concentration of PM_{2.5} (0.13 µg/m³), NO₂ (1.0 µg/m³) and SO₂ (1.6 µg/m³) from industrial emissions was associated with a 19% (95% CI: 17–20%), 21% (95% CI: 19–23%) and 23% (95% CI: 21–24%) increase in the risk of asthma onset in children, respectively. For PM_{2.5} and NO₂, associations were persisting after adjustments for long-term regional PM_{2.5} and traffic-related NO₂ ambient concentration.

Conclusion: Residential exposure to industrial emissions estimated from dispersion modeling was associated with asthma onset in childhood. Importantly, associations were stronger at lower concentrations and independent from those of other sources, thus adding up to the burden of regional and traffic-related air pollution.

1. Introduction

Respiratory diseases represent a significant burden on both the health of the population as well as the healthcare system (GBD, 2017). Asthma is the most prevalent chronic respiratory disease; according to the World Health Organization (WHO), 235 million people worldwide, of all ethnic groups and ages are suffering from asthma (GBD, 2017; WHO, 2019a). It is also the most common chronic disease in children (WHO, 2019b; Global Initiative for Asthma (GINA), 2017). The etiology of asthma is complex, and a number of risk factors have been identified, including exposure to air pollution (Beasley et al., 2015; Gershon et al., 2010; Subbarao et al., 2009a; Subbarao et al., 2009b). Over the past decade several studies have investigated the association between

outdoor air pollution and asthma onset, and the conclusions of most recent review and meta-analysis of 41 studies, including 21 birth cohort studies, support the hypothesis that childhood exposure to air pollution plays a causal role in the development of the disease (Khreis et al., 2017). These evidences mostly derived from studies conducted in urban settings, thus focusing on urban and traffic-related air pollutants, including nitrogen dioxide (NO₂) and particulate matter of aerodynamic diameters ≤ 2.5 µm (PM_{2.5}).

Industries may be significant contributors to air pollution experienced by local populations. Very few cohort studies (Clark et al., 2010; Buteau et al., 2018) have investigated the possible contribution of industrial emissions to the development of asthma. Clark et al. (2010) found that incidence of asthma diagnosis in children aged up to 3–4

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years was associated with the inverse-distance weighted summation of emissions from industrial emitters within a 10 km radius of the residence at birth and during the first year of life. In the other cohort study (Buteau et al., 2018), we found that asthma onset in children was associated with some indicators of exposure to industries developed from the distance between the residence of cohort subjects and the point sources, the tons of air pollutant (PM_{2.5} and sulphur dioxide (SO₂)) emitted by industries, and the percent of time that participants' residence were downwind of industries. Specifically findings indicated that children living in proximity to industries were at greater risk of asthma onset than those who were not, and, that among children exposed the magnitude of the risk of developing the disease increased with the exposure. However, these two cohort studies (Clark et al., 2010; Buteau et al., 2018) were mainly limited by the exposure assessment, which relied on some indicators of exposure. Land-use and terrain as well as meteorology are among factors that contribute to local population's exposure to industrial pollutants.

We sought to consolidate findings from our previous study (Buteau et al., 2018) associating asthma onset in children and to some indicators of exposure to industrial emissions, by refining the exposure assessment. Specifically, we developed estimates of exposure to industrial emissions at residential location from atmospheric dispersion modeling and we assessed the association with childhood asthma onset using the Quebec population-based birth cohort.

2. Methods

2.1. Study population

Detail of the study design has been presented in a previous publication (Buteau et al., 2018). Briefly, we used a population-based birth cohort constructed from the Quebec Integrated Chronic Disease Surveillance System (QICDSS) (Blais et al., 2014). The QICDSS is developed from the linkage of health administrative datasets that include, among others, the registration file from the Quebec medicare system (Régie de l'assurance maladie du Québec, RAMQ), the hospital discharge file, and the fee-for medical service file. In addition, these files include demographic information (e.g., sex, date of birth) as well as the time-varying six-digit postal code of the residence of subjects.

The study population included all children born in Quebec between April 1, 2002, and March 31, 2011. Newborns were identified as new entries of children in the registration file from the RAMQ. Newborns were followed until they developed asthma, died, moved out of Quebec or reached the end of the study period (i.e., March 31, 2011) without developing the disease. Children who moved out of Quebec could not re-enter the cohort, because it was not possible to ascertain whether they developed the disease in the meantime. Censoring was done on the day of loss to follow-up according to the RAMQ data.

2.2. Definitions of asthma onset

New-onset of asthma in children was identified using validated algorithms (Gershon et al., 2010) based on diagnoses and specific health services rendered by physicians both in and out of hospital. Specifically, we classified the new cases of asthma using the following two definitions: 1) at least two medical services (i.e., visits to the emergency room or physician's office) with a diagnosis of asthma within a two-year period, or; 2) one hospital discharge with a primary or secondary diagnosis of asthma. In the former definition, the medical services had to be on different days and, the indexing date was the date of the second service. Asthma diagnoses were identified using ICD-9 code 493 and ICD-10 codes J45-J46.

2.3. Ambient air pollutant concentration from industries estimated from atmospheric dispersion modeling

For each year of the study period and each residential six-digit postal code, we estimated the mean annual ambient concentrations of PM_{2.5}, NO₂, and SO₂ from industrial emissions using the CALMET/CALPUFF modeling system (Scire et al., 2000). These estimates were then linked to individuals of the cohort using their 6-digit residential postal code throughout the follow-up.

A detailed description of the atmospheric dispersion model is presented in Appendix A. Briefly, the modeling system consists of two main components: 1) the meteorological module (CALMET) and; 2) the dispersion module (CALPUFF) that simulates air quality based on industrial emissions. CALMET is a meteorological pre-processor to CALPUFF and it was used to interpolate winds and temperatures using higher-resolution terrain elevation and land-use data and create detailed hourly meteorological fields as well as predict boundary layer parameters such as mixing height. CALPUFF is based on the Lagrangian puff equation, which estimates the growth diffusion and transport of released puffs in the modeling domain. Besides inputting spatially resolved meteorology, emissions from each of the industrial point sources in the Quebec province were input. These emissions data were extracted from the National Pollutant Release Inventory (NPRI) (ECCC, 2016), which is a legislated, publicly-accessible database of pollutant releases, disposals and transfers by Canadian facilities that meet reporting requirements. The NPRI includes data about air pollutant emissions as well as address information about all facilities that manufacture or process any of the NPRI-listed substances. The emission input file for simulation also includes the stack height, stack diameter, exit velocity, exit temperature, emission rates in gram/sec.meter (obtained from NPRI database) and base elevation. It should be noted that no background concentration was considered in the simulation. Because the NPRI reports emission as NO₂ and SO₂ (rather than nitrogen oxides and sulphur oxides) no chemistry were used in the dispersion process.

2.4. Covariates

Age and sex are accepted risk factors for asthma (Beasley et al., 2015; Subbarao et al., 2009a, 2009b) for which information was available from the health databases. Exposure to second hand smoke as well as poor housing conditions (e.g., humid conditions and mold) are among postulated risk factors to the development of the disease (Beasley et al., 2015; Subbarao et al., 2009b), but for which individual-level information was unavailable for all the population. However, as neighbourhood-level socioeconomic status is strongly associated with some of these factors, we used area-based Pampalon's indices of material and social deprivation (Pampalon et al., 2009). The two indices are expressed as quintiles and available for census years at the geographic level of dissemination area (this is the finest spatial resolution at which census data are disseminated). Pampalon's indices for 2001 and 2006 census were used for the 2002–2005 and the 2006–2011 period, respectively. Quintiles were then assigned to cohort subjects through time according to their current residential six-digit postal code.

While SO₂ is mostly emitted by industries, PM_{2.5} are mainly from regional (remote) sources and NO₂ from road-traffic. Therefore, we controlled for long-term measures of regional PM_{2.5} and NO₂ to assess whether long-term exposure to ambient air pollutants from regional and traffic sources may explain the associations between asthma onset and ambient air pollution from industries. For both pollutants, annual ambient air pollutant concentration surfaces were obtained from the Canadian Urban Environmental Health Research Consortium (CANUE; <https://canue.ca/>). Briefly, satellite-based estimates of PM_{2.5} are available across North America for each year of our study period at a

1 km $\times$ 1 km spatial resolution (van Donkelaar, 2015; Boys et al., 2014; DMTI Spatial Inc., 2015). These estimates of ambient concentration of PM_{2.5} were developed using a 1 km optimal estimation aerosol optical depth satellite retrieval, which was related to PM_{2.5} with a chemical transport model (GEOS-Chem CTM) and further adjusted with ground-based PM_{2.5} monitors using a geographically weighted regression.

For ambient NO₂ concentrations, we used outputs from a National land use regression model (Hystad et al., 2011; Weichenthal et al., 2017; DMTI Spatial Inc., 2015), which provide estimates of concentration for each year of our study period and for all six-digit postal code locations in Canada. Briefly, this national land use regression model was developed from the National Air Pollution Surveillance monitoring data, satellite-derived NO₂ estimates, area of industrial land use, road length, and mean summer rain fall and further calibrated by incorporating local-scale variations of NO₂ from vehicle emissions. The model was found to explain 73% of the variation in annual measurements of NO₂ in 2006 and mostly represents levels from road-traffic (Hystad et al., 2011).

2.5. Statistical methods

We used Cox proportional hazards models with age (in days) as the time-scale to assess the relationships between residential exposure to ambient air pollutant concentrations from industries and asthma onset. Separate models were performed for each pollutant using time-varying annual ambient concentrations of air pollutant estimated from dispersion modeling and assigned to each subject throughout the follow-up period according to its current (time-varying) residential postal code.

All models were adjusted for sex and for calendar year (treated as categorical) to account for potential changes in the risk over time. To adjust for potential confounders linked to neighbourhood-level socioeconomic status we used area-based Pampalon's indices of material and social deprivation (Pampalon et al., 2009).

To account for the potential non-linear effect of air pollutant concentration on the hazard of asthma onset, we used restricted cubic splines with three knots located at the 10th, 50th, and 95th percentiles of the distribution (we used the 10th rather than the 5th percentile because the latter was equal to zero for all three pollutants). We visually inspected the fitted exposure–response functions for the pollutant variables and used the AIC (Akaike information criterion; Akaike, 1974) to determine whether the flexible models were consistent with a linearity.

2.6. Sensitivity analyses

To assess whether the estimated associations with air pollutant concentrations from industries were explained by background/regional air pollutant concentration, PM_{2.5} and NO₂ models were further adjusted for ambient regional concentrations from national models (see section 2.3 for a description of the model for PM_{2.5} and NO₂).

We were concerned that adjusting for calendar year may lead to some over-adjustments because of temporal trends in ambient air pollution concentration and industrial emissions across years; therefore in sensitivity analysis we performed Cox models unadjusted for calendar year. As well, diagnosing asthma in very young children, particularly younger than two years old, can be difficult as the primary symptoms of asthma (i.e., wheezing and coughing) may be caused by other conditions. We thus performed a subgroup analysis in children aged two year and older.

We also investigated whether associations differ between residents of Montreal and those of urban and rural environment. More precisely, we used census metropolitan areas (CMAs; area defined as having a total population of at least 100,000 of which 50,000 or more live in the core) to define urban area, whereas all territory outside these CMAs were considered as a rural. We performed separate analysis for residents of the city of Montreal, residents of the six census metropolitan

Table 1

Description of the cohort of children born in the province of Quebec (Canada), 2002–2011.

	n (% male)
Children born in Quebec between 2002 and 2011	722,667 (51%)
Children with asthma diagnosis	66,591 (61%)
Children censored	656,076 (50%)
Because of death	1248 (57%)
Because of relocation	26,421 (52%)
Because the study period ended	628,407 (50%)
Person-years of follow-up	
Sex	
Male	1,600,287
Female	1,569,758
Age-group (year)	
<1	710,146
1–5	2,106,502
6–9	353,396
Pampalon's quintile for material deprivation	
1	623,521
2	636,920
3	610,554
4	590,993
5	611,293
Missing or invalid postal code	96,763
Pampalon's quintile for social deprivation	
1	676,705
2	678,851
3	627,948
4	581,889
5	507,888
Missing or invalid postal code	96,763

areas in Quebec (i.e., greater Montreal, Quebec city, Gatineau, Sherbrooke, Saguenay and Trois-Rivières) and those residing outside CMAs.

Lastly, we assessed the potential confounding by unmeasured secondhand smoking (SHS) using an indirect adjustment method (Steenland and Greenland, 2004), extended for continuous exposure (Villeneuve et al., 2011). A detailed description of the methods is provided in Supplemental Material. Briefly, we used data from the Canadian Community Health Survey (Statistics Canada, 2017), which include the residence location participants and information about smoking among adults who had children, thus allowing us to compute the prevalence of children's exposure to SHS for quintiles of exposures to ambient air pollutant concentration from industries. This information, along with the estimate of association between the development of childhood asthma and exposure to SHS (1.31; 95% CI: 1.23, 1.42; Tinuoye et al., 2013), were used to estimate the potential confounding effect of SHS on the estimated hazard ratios.

3. Results

Table 1 describes the birth cohort that comprised 722,667 children born in Quebec between April 1, 2002, and March 31, 2011. On average, children were followed for 4.4 years, for a total of 3.17 million person-years of observations. Throughout the study period we identified 66,591 incident cases of asthma, most of them occurring before four years of age (Fig. 1).

Table 2 shows the distribution of annual average concentrations of industrial emission-related air pollutants estimated by CALPUFF dispersion modeling, as well as regional related ambient air pollutants, at residences of the cohort. From CALPUFF dispersion modeling, the industrial emission-related annual average concentration at participants' residences was 0.19 $\mu\text{g}/\text{m}^3$ (interquartile range (IQR): 0.13 $\mu\text{g}/\text{m}^3$) for PM_{2.5}, 1.26 $\mu\text{g}/\text{m}^3$ (IQR: 1.06 $\mu\text{g}/\text{m}^3$) for NO₂ and 2.19 $\mu\text{g}/\text{m}^3$ (IQR: 1.61 $\mu\text{g}/\text{m}^3$) for SO₂. The average annual ambient concentration of regional PM_{2.5} at participants' residences was 6.85 $\mu\text{g}/\text{m}^3$ (IQR: 3.85 $\mu\text{g}/\text{m}^3$), whereas for NO₂, it was 11.18 $\mu\text{g}/\text{m}^3$ (IQR: 11.78 $\mu\text{g}/\text{m}^3$).

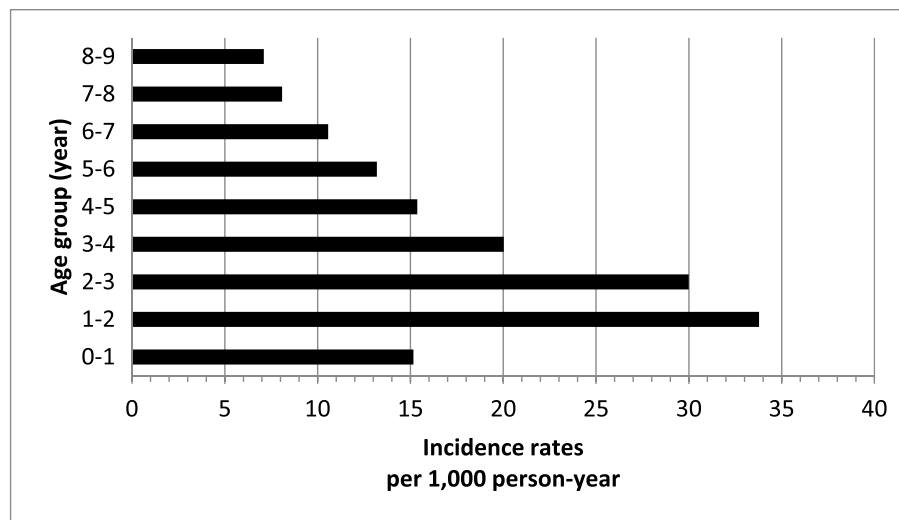


Fig. 1. Incidence rates of asthma according to age of onset in children born in Quebec, Canada, 2002–2011.

It should be noted that the maximum concentration of $PM_{2.5}$ and NO_2 from CALPUFF are greater than from national models; this is due to the fact that our dispersion modeling system provided outputs at a finer spatial scale than the national models (A lower resolution will have a tendency to average out higher concentrations).

Table 3 shows Pearson correlation coefficients (PCC) among air pollutants. Industrial air pollutants were all positively correlated, with stronger correlation observed between industrial SO_2 and $PM_{2.5}$ exposures (PCC = 0.82). Regional ambient $PM_{2.5}$ and NO_2 , estimated from national models, were also highly correlated (PCC = 0.80).

Fig. 2 shows the fitted response functions between the risk of asthma onset and the industrial emission-related annual mean ambient concentration of $PM_{2.5}$, NO_2 and SO_2 from industrial emissions. These response functions depict the hazard of asthma onset relative to those unexposed to industrial emissions. For all pollutants, we concluded that the relationship was nonlinear according to visual inspection of the curves and comparison of the AIC (Supplemental Material Table S2). In all instances, the non-linear positive relationship showed no evidence of a threshold at lower concentration. For all pollutants, the slope of the concentration-response curve was steeper at lower exposure levels and reached a plateau at higher ranges of the exposure distribution.

Table 4 summarizes the results of these figures by tabulating the hazard ratio of asthma onset according to selected changes in the level of exposure to industrial emissions. Specifically, the first column compares the 25th to the 1st percentile to, whereas the second and third columns show the effects comparing the 75th to the 25th percentile and

the 99th to the 75th percentile, respectively. For example, an increase in the annual mean concentration of $PM_{2.5}$ from 0.10 $\mu g/m^3$ (25th percentile) to 0.23 $\mu g/m^3$ (75th percentile) was associated with a 19% (95% CI: 17–20%) increase in the risk of asthma onset. For NO_2 and SO_2 , an increase in the annual mean concentration from the 25th percentile to the 75th percentile (NO_2 : 0.6–1.6 $\mu g/m^3$; SO_2 : 1.2–2.8 $\mu g/m^3$) was associated with an 21% (95% CI: 19–23%) and 23% (95% CI: 21–24%) increased risk of asthma onset, respectively.

After regional ambient $PM_{2.5}$ and NO_2 were adjusted in models, model fit was substantially improved (Supplemental Material Table S2), whereas associations for industrial air pollutant exposure were slightly attenuated at lower concentrations, but virtually not affected at greater concentrations (Table 4 shows hazard ratios for selected changes in exposure; Supplemental Material Figs. S4–5 show concentration-response functions). Importantly, in these models both ambient $PM_{2.5}$ and NO_2 from national models were also associated with asthma onset in children. Specifically, concentration-response functions for regional ambient $PM_{2.5}$ and traffic-related NO_2 in our fully adjusted model were linear with hazard ratios of 1.16 (95% CI: 1.14, 1.18) and 1.06 (95% CI: 1.05, 1.88) per interquartile range (i.e., 3.85 $\mu g/m^3$ for $PM_{2.5}$ and 11.78 $\mu g/m^3$ for NO_2) increase, respectively.

Findings from sensitivity analysis showed that removing calendar year from our main model (Supplemental Material Table S3) as well as restricting the analysis to children aged 2 years and older had marginal influence on the estimated associations (Supplemental Material Table S4). Findings for urban and rural were generally fairly consistent, but

Table 2

Distributions of the mean annual concentrations of $PM_{2.5}$, NO_2 and SO_2 from industrial emissions at children's residence in Quebec (Canada), 2002–2011, estimated from CALPUFF dispersion modeling (first three rows), and of regional ambient $PM_{2.5}$ and NO_2 from national models (bottom two rows).

Air pollutant	Mean annual ambient air pollutant concentration at children's residence										
	Percentiles of the distribution										
	Mean	Standard deviation	Min	1st	10th	25th	50th	75th	90th	99th	Max
Concentration of industrial air pollutant emissions from dispersion models											
PM _{2.5} (in µg/m ³)	0.19	4.18	0	0	0.04	0.10	0.18	0.23	0.31	0.70	20.80
NO ₂ (in µg/m ³)	1.26	20.28	0	0	0.21	0.58	0.91	1.64	2.51	5.40	60.57
SO ₂ (in µg/m ³)	2.19	33.12	0	0	0.43	1.19	1.85	2.80	3.98	8.23	103.19
Concentration of ambient air pollutantsfrom national models											
PM _{2.5} (in µg/m ³)	6.85	39.55	0.95	2.55	3.97	5.00	6.73	8.85	9.633	11.73	17.53
NO ₂ (in µg/m ³)	11.18	133.07	0.54	2.03	3.16	4.87	8.82	16.65	22.63	31.44	58.00

Table 3

Pearson correlation coefficient of the mean annual ambient concentrations of PM_{2.5}, NO₂ and SO₂ from industrial emissions and of regional ambient PM_{2.5} and NO₂, estimated at participants' residence in Quebec (Canada), 2002–2011.

	Industrial-PM _{2.5}	Industrial-NO ₂	Industrial-SO ₂	Regional-PM _{2.5}	Regional-NO ₂
Industrial-PM _{2.5}	1	0.31	0.82	0.12	0.12
Industrial-NO ₂		1	0.51	0.49	0.51
Industrial-SO ₂			1	0.28	0.30
Regional-PM _{2.5}				1	0.80
Regional-NO ₂					1

there were some slight differences depending on the pollutant and the range of exposure (Supplemental Material Tables S5–S6). For Montreal, mean effect estimates were, in general, slightly attenuated as compared to other regions, with wider confidence intervals that often included the null. For exposure to industrial SO₂, effect estimates were greater in rural areas. Our assessment of potential confounding showed that the prevalence of secondhand smoke exposure was fairly constant across the quintiles of exposures to industrial air pollution emissions (Supplemental Material Table S7) and, consistently SHS exposure had

virtually no influence on the estimated associations (Supplemental Material Table S8).

4. Discussion

In this population-based birth cohort study, we assessed the associations between residential exposure to industrial emission-related ambient air pollutants and new-onset of childhood asthma. Our findings for exposure to industrial PM_{2.5}, NO₂ and SO₂ were consistent;

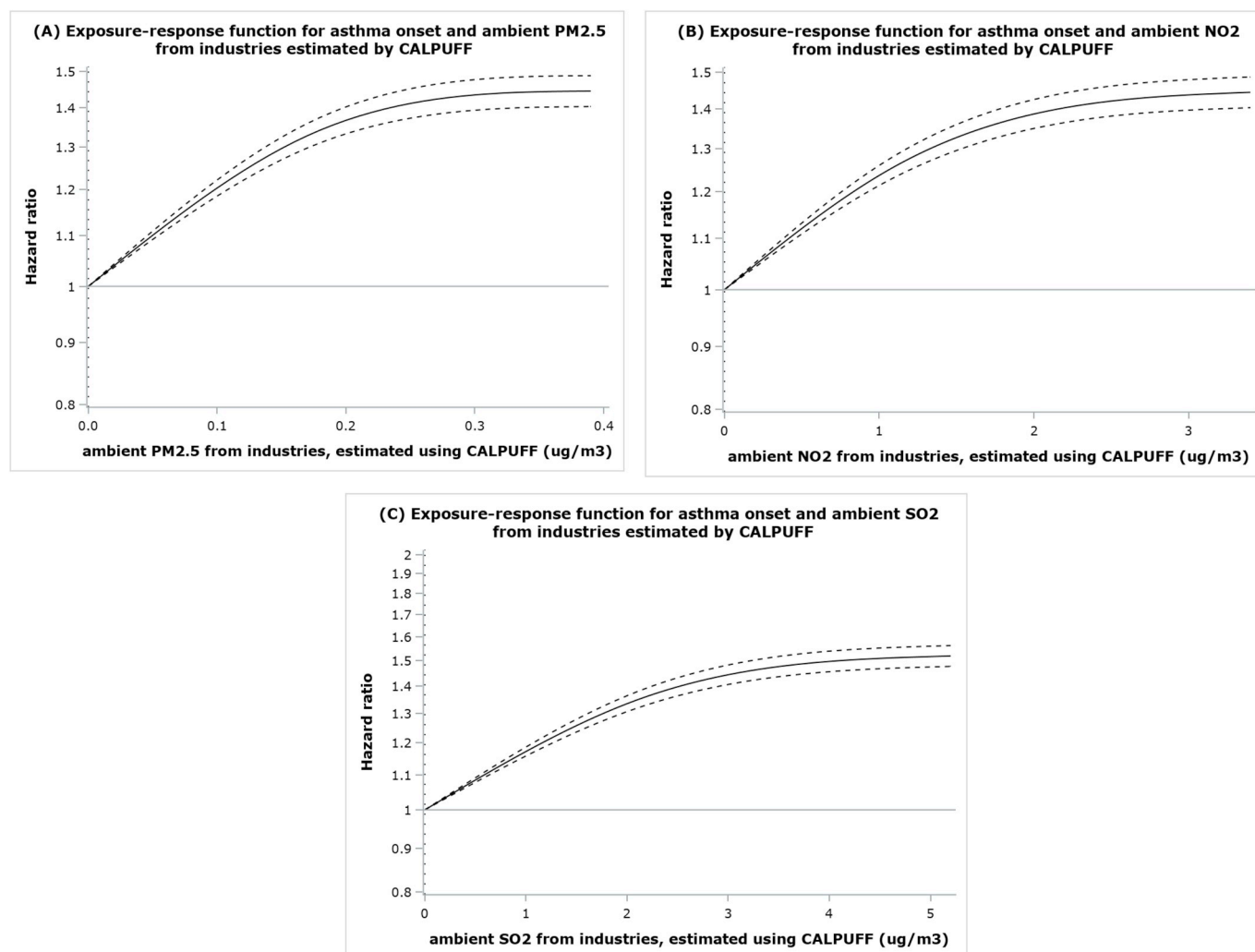


Fig. 2. Adjusted response function between the hazard of asthma onset in children and the annual mean concentration of (A) PM_{2.5}, (B) NO₂, and (C) SO₂ from industrial emissions estimated at participants' residence using dispersion modeling. The horizontal axis represents the mean annual air pollutant concentration (in $\mu\text{g}/\text{m}^3$) from industrial emissions estimated from dispersion modeling. The vertical axis represents the hazard ratio (HR) using children unexposed (annual mean concentration = 0) as the reference (i.e. HR = 1.0). The solid line represents the mean HR from the non-linear function fitted using restricted cubic splines with three knots, with the dashed lines representing the associated 95% confidence interval. All response-functions were estimated from Cox proportional hazards model with age as the time-scale, adjusted for quintiles of material and social deprivation, sex and calendar year. Response-functions are shown up to the 95th percentile of the exposure distribution, which corresponds to the value of the last knot; beyond this last knot the restricted cubic spline function is assumed to be linear.

Table 4

Adjusted hazard ratios and 95% confidence intervals (CI) of asthma onset in children for selected cut-points in the distribution of exposure to the industrial emission-related mean annual pollutant concentrations at children's residences in Quebec (Canada), 2002–2011.

Air pollutant exposure from industrial emissions	Hazard ratio (95% CI)		
	25th vs 1st percentile	75th vs 25th percentile	99th vs 75th percentile
PM_{2.5} from industries			
Main model	1.18 (1.17, 1.20)	1.19 (1.17, 1.20)	1.03 (1.02, 1.05)
Main model, adjusted for regional PM _{2.5}	1.10 (1.08, 1.12)	1.10 (1.09, 1.12)	1.04 (1.02, 1.05)
NO₂ from industries			
Main model	1.13 (1.11, 1.14)	1.21 (1.19, 1.23)	1.08 (1.04, 1.12)
Main model, adjusted for regional NO ₂	1.10 (1.09, 1.12)	1.17 (1.15, 1.20)	1.07 (1.04, 1.10)
SO₂ from industries			
Main model	1.18 (1.16, 1.19)	1.23 (1.21, 1.24)	1.08 (1.05, 1.11)

Main model: Cox proportional hazards model with age as the time-scale, adjusted for quintiles of material and of social deprivation, sex and calendar year. The 1st, 25th, 75th and 99th percentile of pollutant concentrations from industries are: 0, 0.10, 0.23 and 0.70 $\mu\text{g}/\text{m}^3$ for PM_{2.5}; 0, 0.58, 1.64, 5.40 $\mu\text{g}/\text{m}^3$ ce:sup> for NO₂; 0, 1.19, 2.80 and 8.23 $\mu\text{g}/\text{m}^3$ for SO₂.

associations were all positive and non-linear, with concentration-response curve indicating stronger effects at lower concentrations and no discernible threshold. The relatively large increase in risk, despite very low change in industrial air pollution exposure, suggest that industrial point sources, even emitting below standards, may still be hazardous to public health and substantially contribute to the development of asthma in childhood. Importantly, these associations for PM_{2.5} and NO₂ were persisting after controlling for long-term regional ambient PM_{2.5} and traffic-related NO₂ concentration from national models. More precisely, both exposure terms were found to be associated with asthma onset when included simultaneously in our models, thus suggesting that the effects from industrial emissions are independent from those of regional or traffic-related sources. Likely, the spatial resolution of these national air pollution models may not be fine enough to capture the local influence of point sources. This finding has important public health implications; it suggests that the effects and burden of industrial point source emissions adds to that of ambient air pollution estimated from national exposure models.

This study consolidates results from our previous study (Buteau et al., 2018), and adds weight to the evidence linking the development of childhood asthma and ambient air pollution, especially from industrial sources. In our previous study, we found within the same cohort that childhood asthma onset was consistently associated with a number of indicators of exposure developed from the distance, the amount of emissions, and the proportion of time that the residence was downwind of the industries nearby. Herein, we made use of dispersion modeling to refine the exposure assessment and consolidate our previous study findings. Our atmospheric dispersion modeling system allowed us to account for factors including industrial emissions, land-use and terrain, and meteorological information, which likely improved the estimation of exposure to emissions of industries as compared to our previous indicators of exposure. Compared to our previous study (Buteau et al., 2018), the current study showed stronger associations between industrial emission-related ambient air pollution at residential level and children's asthma onset. It is plausible that a refined estimation of population exposure to industrial emissions have contributed in reducing the exposure misclassification.

To the best of our knowledge, this study is the first to quantify the associations between childhood asthma onset and estimations of industrial emission-related ambient pollution at residential level using a dispersion modeling technique. A main strength of this study relates to its population-based design, comprising the entire population of newborns in the Province of Quebec. The substantial length of follow-up combined to the large cohort provided enough statistical power to detect small effects. Another strength relates to the fact that we had information about the full (i.e., 6-digit) residential postal code through time, which allowed us to reconstruct the residential history of children and to assign air pollutant exposure at a relatively fine spatial scale.

Assessing exposure to air pollution from industries is very challenging. A recent review of air pollution studies investigating asthma-related health effects in childhood has identified the exposure assessment as a main limitation (Buteau et al., 2019). In these previous studies, the exposure was mostly ecological and binary, based on an arbitrarily cutoff distance or the community of residence. By accounting for several factors influencing the spatial dispersion of emissions (e.g., tons emitted, wind speed and direction, topography), dispersion modeling technique may substantially enhance the precision in estimating population exposure to industrial emission-related ambient air pollutants, reduce exposure misclassification, and provide a more precise assessment on the adverse effects of industrial point source emissions. One limitation to our exposure estimates is that emissions data used are not based on actual measurements but rather on engineering calculations and self-reporting by the industries. In addition, the NPRI reporting requirements are designed to capture the most significant point-sources, thus some small industrial facilities were likely not included in the NPRI records.

Another limitation relates to the lack of information on participant-level risk factors. Family history of atopy is a main risk factor of asthma onset (Beasley et al., 2015) that we could not account for. Individual socioeconomic status was controlled by using neighbourhood socioeconomic status. However, as neighbourhood socioeconomic status is strongly associated with individual socioeconomic status and behavioural variables (Gan et al., 2012; Janssen et al., 2006), adjusting for neighbourhood socioeconomic status should provide, to some extent, adjustments for unmeasured risk factors. Our sensitivity analysis showed that second-hand smoke exposure had no confounding influence on the estimated associations. This is consistent with our previous study (Buteau et al., 2018), as well as with other studies of adverse health effects of ambient air pollution in Canada (Tétreault et al., 2016; Crouse et al., 2015; Chen et al., 2013; Villeneuve et al., 2011, 2013).

Distinguishing asthma from other respiratory illness (e.g. bronchiolitis) may be challenging in very young children; thus our analysis is subject to some misclassification of asthma onset. The definition used for asthma is the one adopted for surveillance purposes in Canada and also used in some previous cohort studies (Tétreault et al., 2016; Buteau et al., 2018). It has been estimated that this definition has an 89% sensitivity and a 72% specificity in Canadian children aged between 0 and 17 years (Gershon et al., 2010). Our sensitivity analysis showed that results were consistent when restricting to children aged two years and older.

5. Conclusions

In this population-based birth cohort, new-onset of asthma in childhood was significantly associated with exposure at residential location to ambient air pollutant concentration emitted from industrial

point-sources, estimated by dispersion modeling. Importantly, the concentration-response curves showed no evidence of a threshold and stronger effects at lower concentrations, suggesting that even low industrial air emissions may contribute to the development of asthma in childhood. Furthermore, associations for PM_{2.5} and NO₂ from industrial sources were found to be independent from those of ambient air pollutant concentrations estimated from national exposure models. This stresses the importance of better considering industrial point-sources in future air pollution health effects studies and when assessing the burden of diseases from ambient air pollution.

Declaration of competing interest

None declared.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envres.2020.109180>.

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